SLATE PCB — Hand Assembly Parts List

Board: ESP32_PRELIM_DESIGN (66 × 40 mm, 4-layer). Generated from the final schematic after ERC/DRC/parity passed. Tick each line as you place it.

70	6	R8 / R9
parts to solder (36 line items)	need hot air or a hotplate (U2, U3, U4, U6, U7, L1)	fit ONE at most — both = VSYS short

Parts to solder

	Reference(s)	Qty	Description	Part number / supplier	Method
Hot air / hotplate					
☐	U4	1	Li-ion fuel gauge, 8-TDFN-EP 2x2 mm Hardest part. Thin paste on centre pad; check pins 2-3 (VBAT) for bridges to the centre pad.	**MAX17048G+T10** Analog Devices (Maxim) DigiKey · LCSC	HOT AIR
☐	U6	1	Buck-boost converter 3.3 V, VSON-14 3x4 mm Centre pad has open thermal vias: use a little extra paste. Pin 1 = VINA (top-left in silkscreen).	**TPS63020DSJR** Texas Instruments DigiKey · LCSC	HOT AIR
☐	U7	1	Li-ion charger with power path, VQFN-16 3x3 mm Centre pad must be soldered (4 thermal vias).	**BQ24075RGTR** Texas Instruments DigiKey · LCSC	HOT AIR
☐	U3	1	I2S class-D amplifier, TQFN-16 3x3 mm Centre pad to GND (5 vias).	**MAX98357AETE+T** Analog Devices (Maxim) DigiKey · LCSC	HOT AIR
☐	L1	1	Inductor 1.5 uH, 4x4x2.1 mm shielded Terminals are under the body. Also sold direct at coilcraft.com.	**XAL4020-152MEB** Coilcraft DigiKey · LCSC · Maker	HOT AIR
☐	U2	1	ESP32-S3 Wi-Fi/BLE module (16 MB flash, 8 MB PSRAM) Edge pads can be iron-soldered, but the GND pad underneath needs hot air from below or a hotplate. N8R8 also fits.	**ESP32-S3-WROOM-1-N16R8** Espressif DigiKey · LCSC	HOT AIR
Fine pitch (iron + flux, or hot air)					
☐	J5	1	USB-C receptacle, USB 2.0, 16-pin top mount 0.5 mm pitch signal pins: drag-solder with flux, check for bridges. Solder the 4 shell legs last.	**USB4105-GF-A** GCT DigiKey · LCSC	IRON
☐	U1	1	Haptic driver, VSSOP-10 (MSOP) Drag-solder with flux. No centre pad.	**DRV2605LDGSR** Texas Instruments DigiKey · LCSC	IRON
☐	U5	1	USB ESD protection, SOT-23-6 Pins 1, 2 and 5 have vias in the pad: add a little extra solder.	**USBLC6-2SC6** STMicroelectronics DigiKey · LCSC	IRON
Diodes / transistor (iron)					
☐	D3	1	Dual Schottky, common cathode, SOT-23 Orientation fixed by the 3-pad footprint.	**BAT54C,215** Nexperia DigiKey · LCSC	IRON
☐	D6	1	Schottky diode 30 V, SOD-323 Polarity: cathode band toward the right (toward C21 / the EN via).	**BAT54WS-7-F** Diodes Inc. DigiKey · LCSC	IRON
☐	Q1	1	N-channel MOSFET 60 V, SOT-23	**2N7002,215** Nexperia DigiKey · LCSC	IRON

LEDs (iron, check polarity)					
☐	**D1, D2, D5**	3	LED green 0805 (Vf ~2.1 V) Cathode on the LEFT (silkscreen bar). Do not use blue/white: they run from 3.3 V via 1.5 k.	**LTST-C171GKT** Lite-On DigiKey · LCSC	IRON
☐	**D4**	1	LED red 0805 (Vf ~2.0 V) - charging Cathode on the LEFT.	**LTST-C171KRKT** Lite-On DigiKey · LCSC	IRON
Capacitors (iron)					
☐	**C3, C5, C8, C10, C13**	5	0.1 uF 50 V X7R 0805	**CL21B104KBCNNNC** Samsung DigiKey · LCSC	IRON
☐	**C1, C2, C4, C15**	4	1 uF 25 V X7R 0805	**CL21B105KAFNNNE** Samsung DigiKey · LCSC	IRON
☐	**C18, C20**	2	4.7 uF 25 V X5R 0805	**CL21A475KAQNNNE** Samsung DigiKey · LCSC	IRON
☐	**C9, C11, C12, C14, C21**	5	10 uF 25 V X5R 0805 C11 sits right at U6 pins 10/11; C21 is the new EN hold-up cap.	**CL21A106KAYNNNE** Samsung DigiKey · LCSC	IRON
☐	**C6, C7, C16, C17, C19**	5	22 uF 25 V X5R 1206 C6 is now the cap at (205.6, 101.7) - swapped with C5.	**CL31A226KAHNNNE** Samsung DigiKey · LCSC	IRON
Resistors 1% 0805 (iron)					
☐	**R8 OR R9 (one only)**	1	0 R jumper - amp gain select (see gain table) FIT ONLY ONE. Both fitted = VSYS shorted to GND. R8 only = 6 dB, R9 only = 12 dB, neither = 9 dB.	**RC0805JR-070RL** Yageo DigiKey · LCSC	IRON
☐	**R4**	1	33 R	**RC0805FR-0733RL** Yageo DigiKey · LCSC	IRON
☐	**R5, R11, R24, R26**	4	1.5 k LED resistors: labels hidden on silkscreen, all four identical.	**RC0805FR-071K5L** Yageo DigiKey · LCSC	IRON
☐	**R18**	1	2 k (charge current ~445 mA)	**RC0805FR-072KL** Yageo DigiKey · LCSC	IRON
☐	**R17**	1	3.3 k (input limit ~470 mA) Changed from 1.18 k.	**RC0805FR-073K3L** Yageo DigiKey · LCSC	IRON
☐	**R1, R2**	2	4.7 k (I2C pull-ups)	**RC0805FR-074K7L** Yageo DigiKey · LCSC	IRON
☐	**R15, R16**	2	5.1 k (USB-C CC)	**RC0805FR-075K1L** Yageo DigiKey · LCSC	IRON
☐	**R3, R6, R7, R12, R20, R22, R25**	7	10 k	**RC0805FR-0710KL** Yageo DigiKey · LCSC	IRON
☐	**R13, R14**	2	100 k	**RC0805FR-07100KL** Yageo DigiKey · LCSC	IRON

☐	**R21**	1	180 k (U6 feedback)	**RC0805FR-07180KL** Yageo DigiKey · LCSC	IRON
☐	**R10**	1	680 k (amp SD_MODE)	**RC0805FR-07680KL** Yageo DigiKey · LCSC	IRON
☐	**R19, R23**	2	1 M	**RC0805FR-071ML** Yageo DigiKey · LCSC	IRON
Switches (iron)					
☐	**SW1, SW2, SW3**	3	Tactile switch SPST, SMD (C&K KMS2 series) SW1 reset, SW2 boot, SW3 power. Listed as side-actuated: confirm this is the direction you want.	**KMS231GP LFS** C&K DigiKey · LCSC	IRON
Connectors (iron, last)					
☐	**J1**	1	JST PH 2-pin side-entry - battery Check your battery lead polarity against pin 1 = VBAT before plugging in.	**S2B-PH-K-S(LF)(SN)** JST DigiKey · LCSC	IRON
☐	**J3, J4**	2	JST PH 5-pin side-entry - PDM mics	**S5B-PH-K-S(LF)(SN)** JST DigiKey · LCSC	IRON
☐	**LS1**	1	Molex PicoBlade 2-pin right-angle SMD - speaker	**0532610271** Molex DigiKey · LCSC	IRON
☐	**J2**	1	Female header 1x11, 2.54 mm, through-hole - display/SD Tin version PPTC111LFBN-RC also fits.	**PPPC111LFBN-RC** Sullins DigiKey · LCSC	IRON

Not soldered / no part

- **R8 / R9 — amp gain select (MAX98357A GAIN_SLOT)**. **Never fit both:** R8 goes VSYS→GAIN_SLOT and R9 goes GAIN_SLOT→GND, so two 0 Ω parts short VSYS (and the battery) to ground.

Fit	GAIN_SLOT	Gain
R8 = 0 Ω only	VSYS	6 dB
R8 = 100 kΩ only	100 k to VSYS	3 dB
neither	open	9 dB (schematic default)
R9 = 0 Ω only	GND	12 dB
R9 = 100 kΩ only	100 k to GND	15 dB

  The schematic marks both as not fitted (9 dB); if you fit one, update its DNP flag so the BOM matches.
- **H1, H2, H3** — M2 mounting holes: M2 screws + standoffs, nothing to solder.
- **TP1, TP2** — haptic motor (ERM/LRA) wire pads: solder the motor leads directly.
- **GND1, SCL1, SDA1, RX1, TX1, FG_ALRT1** — bare test pads.
- **Mating plugs (optional):** JST PHR-2 (battery), JST PHR-5 ×2 (mics), Molex PicoBlade 51021-0200 (speaker), plus crimp contacts or pre-crimped leads.

Assembly order

1. **Hot air / hotplate first** (paste, ~350 °C air, low flow): U4 → U6 → U7 → U3 → L1. If you have a hotplate, do these plus U2 in one pass.
2. **U2 (ESP32 module)**: hotplate or hot air from below for the GND pad, then touch up the edge pads with an iron.
3. **J5 USB-C**: tack opposite corners, drag-solder the 0.5 mm pins with flux, then the shell legs.
4. **U1, U5**: drag-solder with flux.
5. **Passives by area**: power section (C11, C12, C13, C18, C20, C21, C14, C15, C16, C17, C19, R14, R17–R21, R23) first, then ESP area (C5, C6, R3, C4, R6, R1, R2), then the rest.
6. **D3, D6, Q1, LEDs** — check polarity (see right).
7. **Switches** SW1–SW3 (quick heat; plastic body).
8. **Connectors last**: J1, J3, J4, LS1, J2.
9. Clean flux (IPA), inspect every QFN/DFN edge under magnification.

Polarity / orientation

- **LEDs D1, D2, D4, D5**: cathode on the **left** (silkscreen bar), all four rows the same way.
- **D6** (BAT54WS): cathode band toward the **right** (C21 / EN via side).
- **ICs**: match the pin-1 dot on the part to the pin-1 mark on the silkscreen.
- **J1 battery**: pin 1 = VBAT. JST battery leads are not standardised — check with a meter before connecting.

Before first power

- Meter for shorts: VBUS–GND, VBAT–GND, VSYS–GND, 3.3V–GND.
- First power from a bench supply on J1 at 3.7 V, 200 mA limit. Press SW3: 3.3V rail should come up (~3.28 V) while held.
- Then USB: 3.3V should stay on without holding SW3 (VBUS → D6 → U6 EN).

> **Firmware must:** drive IO1 (PWR_HOLD) high at the start of `setup()`; enable the internal pull-up on IO2 (PWR_SENSE). Power-off takes ~1–2 s after PWR_HOLD goes low (C21 hold-up).

Notes on the part choices

- Capacitors are rated well above their rail (5 V max) to limit DC-bias loss. Any X5R/X7R part of the same size, value and voltage can substitute.
- Resistors: any 1% 0805 thick-film part works; Yageo RC0805FR series shown.
- LEDs must have Vf ≤ ~2.2 V (red, green, yellow, orange). The status LEDs run from 3.3 V through 1.5 kΩ (~0.7–0.9 mA).
- DigiKey / LCSC links open a part-number search, so you see live stock and alternatives. Confirm stock and package before ordering; buy 2–5 spares of each passive.
- Charger settings: R18 = 2 kΩ → ~445 mA charge (0.37C for 1200 mAh); R17 = 3.3 kΩ → ~470 mA input limit (USB 2.0). R20 fixes TS (no battery temperature sensing — use a pack with a protection board).